Syllabus Details

Syllabus ID:	13165
Syllabus Name:	**Java Web Application Development_Phát triển ứng dụng Java web**
Course Name English:	**Java Web Application Development**
Subject Code:	**PRJ301**
Learning-Teaching Method:	Project based learning,Active learning,Social Constructivism,In-class lecture
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 85' practical exam + 102.6h self-study
Pre-Requisite:	DBI202, PRO192
Description:	By the end of this course Students will be able to: a) Knowledge: (what will students know?) • Understand the core technologies of Java web programming: - Servlet and JSP - Scope of sharing state (session, application, request,page) • Know how to develop and deploy your own websites using Java • Understand and be able to apply MVC architecture for the web - Learning how to apply JPA in websites using Java • Learning how to apply AI in your own website using Java b) Skills: (what will students be able to do?) • Basic Web application development applying MVC Design Pattern using Servlet/Filter as Controller • Creating a simple Web Application demo using JPA
StudentTasks:	- Attend at least 80% of contact hours in order to be accepted to the final examination - Actively participate in class activities - Fulfil tasks given by intructor after class - Use their own laptop in class only for learning purpose - Read the textbook in advance - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	Web server application :Tomcat 9 or higher IDE: Netbeans 13 or higher (as defined by the subject leader) Java EE: 1.8 or higher MS SQL Server 2019 or higher
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	932/QĐ-ĐHFPT dated 08/22/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/22/2025

6 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online
1	Professional Java® for Web Applications	Nicholas S. Williams	Wrox		1st edition	978-1118656464	True	False	True
2	Article 1: http://docs.oracle.com/javase/tutorial/JDBC						False	False	True
3	Article 2: https://docs.oracle.com/cd/B14099_19/web.1012/b14017/filters.htm						False	False	True

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online
4	Session 3 Guide to Using Prompt Engineering Effectively with Java Web						False	False	False
5	Session 8 AI_Netbeans24						False	False	False
6	Session 15 Using a Database Query Agent in Java Web Applications						False	False	False

9 LO(s)

No.	CLO Name	CLO Details
1	CLO1	understand the basic web application structure and be able to deploy Web application
2	CLO2	understand and be able to work with the basic features of Java web application
3	CLO3	be able to work with JDBC
4	CLO4	be able to use different languages in JSP
5	CLO5	understand and be able to work with other server side objects in Java Web Application
6	CLO6	understand and be able to work with simple MVC Architecture
7	CLO7	be able to combine web front-end & back-end in the Java web application.
8	CLO8	be able to use JPA in the Java web application.
9	CLO9	be able to use AI in the Java web application.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download
1	Introduction to java web application + The core and basic of Java web server technologies + Web design vs server technologies	Offline	LO1	IT	Textbook: Chapter 1 & 2	PRJ301
2	Setup Environment : JDK: 1.8 or higher Servlet container: Tomcat 10 or higher Intergrate Netbeans 13 with the web container.	Offline	LO1	TU	Textbook: Chapter 1 & 2	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download
3	Creating/Building the first application: Learn to create Servlet Guide to Using Prompt Engineering Effectively with Java Web	Offline	LO1; LO08	TU	Textbook: Chapter 1 & 2	
4	Learn to create JSP Deploy the web application	Offline	LO1	TU	Textbook: Chapter 1 & 2	
5	Wiritting the First Servlet Create a servlet + Create a servlet class + Configure servlet for deployment	Offline	LO2	TU	Textbook: Chapter 3	
6	Work with request and response objects + Understanding DoGet and DoPost	Offline	LO2	TU	Textbook: Chapter 3	
7	+ Using parameter to accept form submition & query string + Response text content to client	Offline	LO2	TU	Textbook: Chapter 3	
8	Using AI in Netbeans24	Offline	LO1	TU	Textbook: Chapter 3	
9	Practice Classwork 1	Offline	LO2	TU	Textbook: Chapter 3	
10	Practice Classwork 1	Offline	LO2	TU	Textbook: Chapter 3	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download
11	Web Application Interacting with Database + JDBC Introduction + Types of JDBC Drivers	Offline	LO3	IT	http://docs.oracle.com/javase/tutorial/jdbc/index.html	
12	Web Application Interacting with Database + JDBC Introduction + Types of JDBC Drivers	Offline	LO3	TU	http://docs.oracle.com/javase/tutorial/jdbc/index.html	
13	+ JDBC Basics: Processing SQL Statements	Offline	LO3	TU	http://docs.oracle.com/javase/tutorial/jdbc/index.html	
14	+ JDBC Basics: Processing SQL Statements	Offline	LO3	TU	http://docs.oracle.com/javase/tutorial/jdbc/index.html	
15	JDBC Basics: Processing SQL Statements Use a Database Query Agent in Java Web Applications	Offline	LO3; LO08	TU	http://docs.oracle.com/javase/tutorial/jdbc/index.html	
16	Implement CRUD application using MS SQL Server 2019 or version higher	Offline	LO3	TU	http://docs.oracle.com/javase/tutorial/jdbc/index.html	
17	Implement CRUD application using MS SQL Server 2019 or version higher	Offline	LO3	TU	http://docs.oracle.com/javase/tutorial/jdbc/index.html	
18	Implement CRUD application using MS SQL Server 2019 or version higher	Offline	LO3	TU	http://docs.oracle.com/javase/tutorial/jdbc/index.html	
19	Practice Classwork 2	Offline	LO3	TU	http://docs.oracle.com/javase/tutorial/jdbc/index.html	
20	Practice Classwork 2	Offline	LO3	TU	http://docs.oracle.com/javase/tutorial/jdbc/index.html	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download
21	Using JSP to display content Create JSP? + Why use JSP? 3 Ways to use JSP + Only HTML in a JSP file + Only JSP code in a JSP file	Offline	LO2	TU	Textbook: Chapter 4	
22	+A mixture of JSP code with HTML text in A JSP file Using the Implicit variables in A JSP	Offline	LO2	TU	Textbook: Chapter 4	
23	Configuring JSP properties in the web.xml	Offline	LO2	TU	Textbook: Chapter 4	
24	Combining JSP and Servlet	Offline	LO2	TU	Textbook: Chapter 4	
25	Practice Classwork 3	Offline	LO2	TU	Textbook: Chapter 4	
26	Practice Classwork 3	Offline	LO2	TU	Textbook: Chapter 4	
27	Introduction to MVC Architecture + General of Java Application Design	Offline	LO6	IT	Textbook: Chapter 4	
28	Introduction to MVC Architecture + General of Java Application Design	Offline	LO6	IT	Textbook: Chapter 4	
29	+ MVC Pattern + How to apply MVC in Java Web Application	Offline	LO6	TU	Textbook: Chapter 4	
30	+ MVC Pattern + How to apply MVC in Java Web Application	Offline	LO6	TU	Textbook: Chapter 4	
31	Practice	Offline	LO6	TU	Textbook: Chapter 4	
32	Practice	Offline	LO6	TU	Textbook: Chapter 4	
33	Maintaining state using session and cookie + Explain why session is nessesary + How session and cookie work	Offline	LO2	TU	Textbook: Chapter 5	
34	+ Store and Retreive data to session and cookie	Offline	LO2	TU	Textbook: Chapter 5	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download
35	Practice Classwork 4	Offline	LO2	TU	Textbook: Chapter 5	
36	Practice Classwork 4	Offline	LO2	TU	Textbook: Chapter 5	
37	Expression Language What and Why use Expression Language How to write with the EL syntax + Immediate evaluation , [ ] operator,array, Arithmetic EL,Relational EL, Static Field	Offline	LO4	TU	Textbook: Chapter 6	
38	+ EL functions + Collections How to use scoped variables in EL expressions + pageContext	Offline	LO4	TU	Textbook: Chapter 6	
39	+ sessionScope + applicationScope Implicit Variables in EL	Offline	LO4	TU	Textbook: Chapter 6	
40	Practice Classwork 5	Offline	LO9	TU	Textbook: Chapter 6	
41	Using JPA + Guide to work with persistance connect to DB	Offline	LO9	TU	Textbook: Chapter 19	
44	Practice Classwork 6	Offline	LO9	TU	Textbook: Chapter 19	
45	Using Custom tag + Guide to create a simple Web Application demo using JPA	Offline	LO9	TU	Textbook: Chapter 19	
48	Practice Classwork 7	Offline	LO4	TU	Textbook: Chapter 19	
49	Introduction to Filter + Understand the purpose of Filter	Offline	LO5	TU	Textbook: Chapter 9	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download
50	+ Create, Declaring and Mapping Filter	Offline	LO5	TU	Textbook: Chapter 9	
51	Forward and Redirect to another servlet Using init parameters Building a Coding AI Agent in a Java Web Application (Servlet-Based)	Offline	LO5; LO8	TU	Textbook: Chapter 9	
52	Practice Classwork 8	Offline	LO5	TU	Textbook: Chapter 9	
53	Introduction to Event Listener + ServletContextEvent + HttpSessionEvent + ServletRequestEvent	Offline	LO5	TU	https://docs.oracle.com/cd/B14099_19/web.1012/b14017/filters.htm	
54	+ ServletContext AttributeEvent + HttpSessionBindingEvent + ServletRequestAttributeEvent	Offline	LO5	TU	https://docs.oracle.com/cd/B14099_19/web.1012/b14017/filters.htm	
55	Assignment Evaluation Course Summary for Final Exam	Offline	LO1, LO2, LO3, LO4, LO5,LO6,LO7	TU		
56	Assignment Evaluation Course Summary for Final Exam	Offline	LO1, LO2, LO3, LO4, LO5,LO6,LO7	TU		
57	Assignment Evaluation Course Summary for Final Exam	Offline	LO1, LO2, LO3, LO4, LO5,LO6,LO7	TU		
58	Assignment Evaluation Course Summary for Final Exam	Offline	LO1, LO2, LO3, LO4, LO5,LO6,LO7	TU		
59	Assignment Evaluation Course Summary for Final Exam	Offline	LO1, LO2, LO3, LO4, LO5,LO6,LO7	TU		
60	Assignment Evaluation Course Summary for Final Exam	Offline	LO1, LO2, LO3, LO4, LO5,LO6,LO7	TU		

0 Constructive question(s)

7 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	1	30.0%	>0	Option 1: 5 weeks; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO7, LO8, LO9	Option 1: Project; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 1; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Front end , back end development	-Teacher gives a requirements -Student implements a web application based on the requirements. The application must consider the aspects of uses (not only back-end but also front-end) - Follow assignment evaluation rubric to grade assignment	
2	Practical Exam	on-going	1	30.0%	>0	Option 1: 85 minutes; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO01, LO02, LO03, LO04, LO05, LO06,	Option 1: Practical Exam; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 3; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Basic features of Java Web		
3	Progress Test 1	on-going	1	5.0%	>0	Option 1: 30 mins; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO1, LO2	Option 1: Multiple Choice; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 20; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Basic features of Java Web		
4	Progress Test 2	on-going	1	5.0%	>0	Option 1: 30 mins; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	s	Option 1: Multiple Choice; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 20; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Basic features of Java Web		
5	Workshop 1	on-going	1	5.0%	>0	Option 1: 1 week; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO1, LO2	Option 1: Home work; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 1; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Basic features of Java Web	-Teacher gives a short requirements -Student implements a web application based on the requirements	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
6	Workshop 2	on-going	1	5.0%	>0	Option 1: 1 week; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO3, LO4	Option 1: Home work; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 1; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Basic features of Java Web	-Teacher gives a short requirements -Student implements a web application based on the requirements	
7	Final Exam	Final exam	1	20.0%	4	60 minute	LO001, LO2, LO03, LO04, LO05, LO06	Multiple Choice	50	Basic features of Java Web		